APPLICATIONS, RESOURCES, AND DECISION GUIDELINE

A. Downstream applications

(1) Intelligent transportation system. The accuracy of spatiotemporal data is crucial for Intelligent Transportation Systems (ITS) functions like traffic prediction, network optimization, and incident early warnings [1]. Tensor decomposition-based methods are highly effective at modeling the spatiotemporal characteristics of traffic data and addressing data loss issues caused by sensor malfunctions or communication interruptions [2]. For example, Tan *et al.* [3] proposed a dynamic tensor imputation algorithm leveraging multi-modal low-rank properties and pattern similarity modeling, significantly enhancing traffic prediction accuracy under missing data conditions. Tang *et al.* [4] introduced a tensor decomposition approach for signalized intersections, modeling traffic flow distribution similarity using sparsely sampled vehicle trajectory data to provide a novel modeling perspective for traffic state awareness in vehicle-infrastructure cooperative environments. Furthermore, Dong *et al.* [5] presented a Laplacian-integrated method combining graph convolutional networks with tensor imputation for traffic prediction with missing data in intercity highway networks. This approach utilizes graph-based similarity modeling to enhance the accuracy of traffic state prediction for large-scale highway networks, even in the presence of partial input data loss and sparse detector distribution, thereby providing an effective and robust solution.

(2) Environment monitoring. Accurate environmental monitoring and trajectory data are vital for functions such as pollutant tracking, disaster early warning, and ecological preservation. This data, including air quality indices, meteorological elements, and maritime trajectories, forms the basis for environmental assessments [6]. However, environmental data often has gaps due to sensor failures, extreme weather, or human error, while trajectory data can be discontinuous from sparse sampling or device drift. To address these challenges, researchers have proposed tensor decomposition-based methods for imputing missing environmental data. The imputed data are then applied to various downstream environmental monitoring tasks, significantly improving the accuracy and reliability of environmental state assessments. For example, Xu *et al.* [7] utilized tensor decomposition and sequence-based correlation modeling to extract latent information from diverse heterogeneous data sources. The imputed data subsequently served as input for a hybrid prediction model for carbon emissions, leading to more accurate forecasts. Qi *et al.* [8] introduced a three-dimensional tensor method specifically for port ship carbon emissions. This approach leverages sub-tensor correlation modeling to decompose the tensor into multiple sub-vectors, enabling multi-dimensional feature extraction and missing data imputation for ship carbon emissions. This provides crucial data support and a decision-making basis for developing energy-saving and emission-reduction measures for ships, thereby promoting green and sustainable development in the shipping industry.

(3) Climate science. In climate science, the integrity and accuracy of spatiotemporal data are essential for climate pattern discovery, extreme weather event prediction, carbon cycle simulation, and climate change impact assessment [9]. However, climate data frequently suffer from missing values due to factors like sparse observation stations, equipment malfunctions, or data transmission interruptions [10]. Tensor decomposition-based techniques for missing data imputation are highly effective in processing high-dimensional, heterogeneous spatiotemporal climate data. These techniques can recover the intrinsic structure and dynamics of such datasets, thereby providing more complete and reliable inputs for climate models. This is crucial for improving the accuracy of climate predictions, gaining deeper insights into the climate system, and formulating strategies to address climate change. For example, Xu *et al.* [11] proposed a supervised tensor decomposition-based multi-task learning method for climate data imputation and prediction. This method encodes climate data as a third-order tensor through feature-based similarity modeling, thereby extracting interpretable spatial and temporal latent factors. Its effectiveness has been validated using global-scale climate datasets. To address the pervasive heterogeneity in climate data, Li *et al.* [12] introduced an improved tensor feature analysis method. This approach involves dividing and reconstructing heterogeneous data into sub-tensors and employing tensor decomposition for feature analysis, effectively mitigating traditional biases. Additionally, it can reveal latitudinal changes in temperature gradients and anomalies in air temperature, significantly enhancing data approximation accuracy in complex regions and during climate anomaly events.

(4) Other applications. Beyond intelligent transportation, environmental monitoring, and climate science, tensor decomposition-based spatiotemporal data imputation has vast applications across diverse fields [13]. In the domain of shared mobility demand prediction, for instance, Yan *et al.* [14] proposed HT-STNet, which combines a tensor decomposition model with an LSTM-based sequence similarity modeling approach. This effectively addresses the challenges of modeling high-dimensional, sparse spatiotemporal data. Its capabilities in dynamic sparsity awareness, multi-scale dimensionality reduction, and lightweight modeling offer an efficient and robust solution for tasks like shared mobility demand forecasting. In the context of energy systems, Madbhavi *et al.* [15] introduced a novel state imputation framework based on block tensor imputation. The primary objective is to capture the spatiotemporal characteristics of power system states through tensor modeling and optimize the imputation process by incorporating physical constraints. Moreover, in the field of medical and health monitoring, Cong *et al.* [16] employed tensor decomposition to impute electroencephalogram (EEG) signals and extract signal features. This approach assists in uncovering physiologically significant patterns from high-dimensional, sparse EEG signals, thereby providing a powerful tool for medical diagnosis, brain-computer interfaces, and cognitive science.

B. Code and datasets

Tensor decomposition-based methods have emerged as a cornerstone for research and applications in spatiotemporal

TABLE S1: Links to the publicly available dataset

Data Classification	Dataset Name	Data Type	Sampling Rate	Data Dimension	Dataset Links
Time-dynamic, spatial-static (Node*Day*Time)	PeMS dataset	Traffic flow	5 minutes	358*91*288 (PEMS03) 307*59*288 (PEMS04) 170*62*288 (PEMS08)	http://pems.dot.ca.gov/
	GTL dataset	Traffic flow/speed/ occupancy	5 minutes	26*31*288	https://gtl.inrialpes.fr/
	Hangzhou dataset	Metro inflow/outflow	10 minutes	81*25*144	https://tianchi.aliyun.com/.../information
	Hangzhou metro passenger flow dataset	Metro inflow/outflow	10 minutes	81*25*144	https://zenodo.org/record/3145404
	Portland	Traffic flow/speed/ occupancy	5 minutes	156*31*288	https://new.portal.its.pdx.edu/?/home
	Seattle	Traffic speed	5 minutes	323*365*288	https://github.com/zhiyongc/Seattle-Loop-Data
	Shenzhen Metro Data	Metro inflow/outflow	10 minutes	172*31*144	https://github.com/yongmeizhao/smartcard-data/blob/main/data.rar
	METR-LA	Traffic speed	5 minutes	207*121*288	https://paperswithcode.com/dataset/metr-la
	GuangZhou traffic speed dataset	Traffic speed	10 minutes	214*61*144	http://www.openits.cn/openData2/792.jhtml
	Birmingham parking dataset	Parking occupancy	30 minutes	30*77*18	https://archive.ics.uci.edu/parking+birmingham
	Bond-Fund holding EPS Fundamentals	Bond excess returns & chars	1 month	8759*206*29	https://sethpruitt.net/2022/03/29/reconciling-trace-bond-returns/
	ShenZhen traffic speed dataset	Traffic speed	15 minutes	156*31*96	https://github.com/lehaifeng/T-GCN/tree/master/data
	Xi'an traffic speed data	Traffic speed	10 minutes	500*30*144	sts.didiglobal.com
	PM10 (London)	Air Quality	Hourly	72*90*24	https://zenodo.org/records/4531304
	Beijing Transportation Smart Card data	Metro inflow/outflow	10 minutes	276*31*144	https://github.com/yongmeizhao/smartcard-data/blob/main/data.rar
Time-dynamic, spatial-dynamic (Node*Time*Attribute)	Xi'an trajectory (Didi)	Vehicle Trajectory	1 minute	500*1440*3	https://sts.didiglobal.com/
	Air track	Flight trajectory	1 minute	1000*1440*5	https://zh.flightaware.com/
	US Coastal AIS	Vessel Trajectory	1 minute	2000*1440*4	https://marinecadastre.gov/accessais/

data imputation. With increasing theoretical advancements and practical demands, a wealth of high-quality public datasets and open-source code resources has become available. These resources provide robust support for algorithm validation, performance comparison, and cross-domain transfer. To systematically organize the currently available data resources and implementation tools, this survey compiles representative public datasets alongside their code implementations, aiming to offer a unified reference framework for future research. Presently, datasets and codes suitable for tensor decomposition-based spatiotemporal data imputation tasks encompass various application scenarios, including traffic flow, subway passenger flow, parking status, financial data, and trajectory information. Based on whether their spatial structure changes over time, these datasets can be categorized into two main types: time-dynamic, spatial-static, and time-dynamic, spatial-dynamic, as illustrated in Tab. S1 and S2.

C. Comparative performance synthesis

This paper provides a systematic classification of representative works in tensor imputation through the lens of spatiotemporal effect modeling. It summarizes the advantages and limitations of various methods, alongside their applicability across different missing patterns (RM/BM/TM/MM), offering a strategic reference for model selection in diverse data scenarios. As detailed in Table S3, we establish a comparative framework based on the triad of "spatiotemporal effect—methodological category—missing pattern". Methods that model spatiotemporal correlation (SC) often rely on static regularization terms, such as low-rank decomposition, tensor nuclear norm (TNN), graph Laplacian regularization, or fixed convolutional kernels. By leveraging local continuity and topological constraints to assist the optimization of the objective function, these methods are generally more robust in scenarios involving RM and BM. However, when data are characterized by TM patterns that cause discontinuities along the time axis, the effective range of information propagation for these static regularizers becomes significantly restricted, which in turn can lead to performance degradation. In contrast, methods that characterize spatiotemporal heterogeneity (SH) employ dynamic and adaptive distribution or representation learning mechanisms (e.g., fast adaptation in meta-learning, generative priors) to mitigate the systematic bias introduced by the global homogeneity assumption. Consequently, they are more likely to maintain robustness under TM and certain MM conditions. However, their efficacy often depends on factors such as the scale of the training data, the appropriateness of the task formulation, and the quality of multi-source data alignment. Approaches based on geographical similarity (GS) typically leverage external semantic or functional contexts (e.g., POI, land use, road hierarchy) to establish cross-regional

TABLE S2: Links to publicly available code

Type	Method	Code Links	Platform
Tensor decomposition-based methods	LRTC-3DST [17]	https://github.com/HaoShu2000/LRTC-3DST	MATLAB
	Meta-TD [18]	https://doi.org/10.6084/m9.figshare.25997134	Python
	BGCP, BTTF, BATF, BTMF, LRTC-TNN [19], [20]	https://transdim.github.io/	Python
	GT-TDI [21]	https://github.com/KP-Zhang/GT-TDI	Python
	CP-ALS-QR [22]	https://github.com/rlminste/CP-ALS-QR	MATLAB
	Hg-TD [23]	https://figshare.com/s/17b6beeb...	Python
Tensor nuclear norm-based methods	MVLR [24]	https://github.com/heyang49/MVLR	Python
	CJSR [25]	https://github.com/xbchen82/CJSR	MATLAB
	LRTC-TMCP [26]	https://github.com/peterchen96/LRTC-TMCP	Python
	ST-LRTC [27]	https://doi.org/10.6084/m9.figshare.20289078	Python
	STH-LRTC [28]	https://github.com/mcgill-smart-transport/traffic_state_estimation	MATLAB
	tensor4kriging [29]	https://github.com/tongnie/tensor4kriging	Python
	LRTC_DeepPnP [30]	https://github.com/peterchen96/LRTC_DeepPnP	Python
	RTC-GTNLN [31]	https://github.com/HaoShu2000/RTC-GTNLN	MATLAB
	STTC-L0-NN [32]	https://github.com/TCCofWANG/STTC-L0...	Python
	ImputeFormer [33]	https://github.com/tongnie/ImputeFormer	Python

associations. This enhancement improves model performance in MM scenarios where observations are extremely sparse or physical topological constraints become unreliable. Nevertheless, the robustness of these methods is more susceptible to issues such as insufficient coverage and noise in the external data.

D. Method selection decision guidelines

To transform the theoretical classification system into an actionable implementation guide, we have integrated the modeling mechanisms and their respective limitations into a priority-based decision framework (Fig. S1). This framework is designed as a "bottleneck-driven" heuristic logic, aimed at identifying an appropriate modeling path based on the primary source of imputation difficulty.

(1) Path selection based on modeling bottlenecks

Users should first evaluate data characteristics to identify the dominant modeling bottleneck:

• **Path A: spatiotemporal heterogeneity (SH) modeling.** This path should be prioritized when the data exhibit strong non-stationarity and local variation, significantly violating global homogeneity assumptions. While generative and multi-view methods are powerful for pattern recovery under extreme sparsity, they require cautious management regarding training instability and the risk of violating physical consistency.

• **Path B: geographical similarity (GS) modeling.** This is the preferred primary path when the intrinsic structural information of the tensor is insufficient (e.g., in scenarios with a high proportion of mixed missing (MM)), but rich and informative external spatial priors (such as POI or road levels) are available. The efficacy of this path is closely tied to the representation quality of external data and its semantic alignment precision with the target variables.

• **Path C: spatiotemporal correlation (SC) modeling.** When the tensor possesses significant intrinsic dependencies—such as mathematical smoothness, periodicity, or clear network topology—this path offers the most efficient and interpretable solution. However, static SC regularization terms may experience substantial performance degradation in long-term missing (TM) scenarios.

(2) Implementation strategy and hybrid extensions

The framework emphasizes that these paths are not mutually exclusive. For complex spatiotemporal datasets exhibiting both heterogeneity and structured dependencies, a "priority-first" hybrid strategy is recommended. In such cases, the core bottleneck should be addressed via the corresponding primary path first (e.g., utilizing SH to capture non-stationarity), followed by the introduction of secondary priors (e.g., GS-based contextual constraints) to further refine the solution space and enhance cross-domain generalization. Consequently, the imputation task shifts from pure algorithmic tuning toward "knowledge engineering", where structured prior knowledge is systematically embedded within the tensor decomposition process.

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TABLE S3: Comparative synthesis of representative tensor-based spatiotemporal imputation methods

Effect	Model	Method	External data required	RM	BM	TM	MM	Strengths	Limitations
SC	Feature-based modeling	CRS [34], MRTC [35], MTC [36], TGMC [37], CP-wALS [38], BNPTD [38], LRTC-SCAD [30], TNN-HTV [40], CTMTR [41], CJSR [42], LTRR-NLS [43], POD-DEIM [44], GLOSS [45], Reg. LRTC [46]	No	○	✓	○	○	Simple, interpretable; strong for periodic data.	Fails under strong non-stationarity; sensitive to noise.
	Physics-based modeling	NT-DPTC [47], HRST-LR [48], GCLSNL [49], FDPTC [50], ST-LRST [51], LRTC-3DST [52], NTF [8], RTTC [53], LCR [54], RTLRR [55], GLOSS [56]	Yes	✓	✓	○	○	High interpretability; robust spatial smoothness.	Requires rigid, accurate physical/structural priors.
	Sequential-based modeling	STLSTM [57], RegTensor [58], LTP [59], NTC [60], LSTM-GL-ReMF [61]	No	○	○	✓	✓	Captures nonlinear temporal evolution (e.g., LSTMs).	High training cost; potential for instability.
	Graph-based modeling	GT-SVD [62], GCRINT [63]	Yes	✓	✓	○	○	Superior for explicit network topologies (e.g., roads).	Sensitivity to graph quality and misspecification.
SH	Meta-learning based modeling	Meta-TD [18]	Yes	✓	✓	✓	○	Adaptive spatial weighting; good for non-stationarity.	Highly sensitive to task/meta-knowledge quality.
	Multi-view based modeling	HAN [64], MATCN [65], SMV-NMF [66]	Yes	✓	✓	✓	○	Fuses diverse perspectives to reduce single-view bias.	Depends on cross-view alignment and consistency.
	Generative-based modeling	ST-GAIN [67], DATC [68], CA-GAN [69]	Optional	○	○	✓	✓	Synthesizes patterns under extreme sparsity (e.g., GAN).	May generate implausible values without strong constraints.
GS	Context-based modeling	CARTE [70], [71], NR-cNTF [72], STDI [73]	Yes	○	✓	✓	✓	Enables knowledge transfer via external semantics.	Reliance on high-quality external multi-source data.
	Encoding-based modeling	GT-TDI [21], TRTF [74]	Yes	○	✓	✓	✓	Captures functional consistency via embeddings.	Performance limited by encoder representation quality.
	Dynamic graph-based modeling	ST-LRTC [75]	Yes	✓	✓	✓	✓	Adapts to evolving topological similarities.	High computational overhead for graph updates.

Note: ✓ = Recommended; ○ = Conditional; × = Not Recommended.

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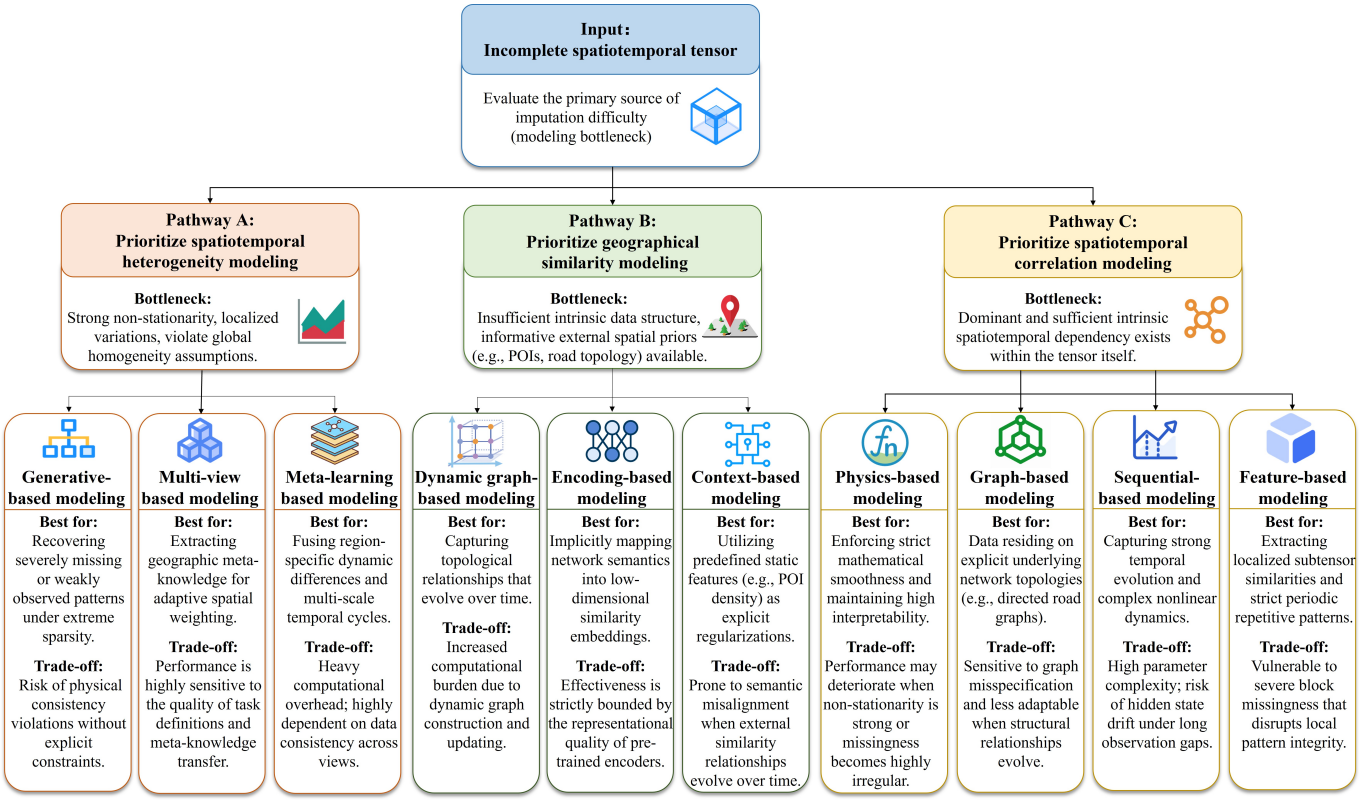

Fig. S1: Priority-based method decision flowchart.

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